Summary in Graph

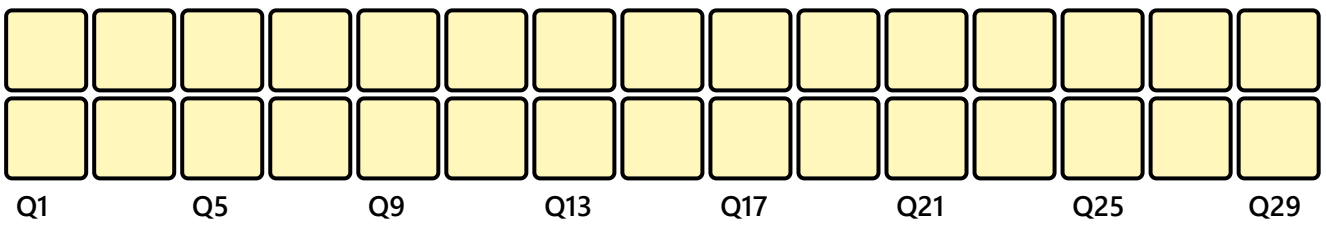
Exam Summary (GO Classes CS Test Series 2026 | Operating Systems | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (20 + 10)
Total Marks:	40 (20 + 20)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (41) out of 409 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

```
Void main() {
    int a = 0;
    for (int i = 0; i < 4; i++) {
        if (fork() == 0) {
            a++;
            return ;
        } else {
            a++;
        }
    }
}
```

The number of possible values(excluding initial value 0) of 'a' after complete execution of the main function?

A. 1
B. 2
C. 4
D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A system shares 12 resources of the same type between 3 processes P1, P2 and P3. The current allocation and maximum requirements of the processes are shown below:

Process	Current Allocation	Maximum Requirement
$P1$	3	4
$P2$	4	9
$P3$	4	8

Which of the following best describes the current state of the system?

A. Safe, Deadlocked
B. Safe, Not Deadlocked
C. Not Safe, Deadlocked
D. Not Safe, Not Deadlocked

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A system is composed of four processes, $\{\mathrm{P1}, \mathrm{P2}, \mathrm{P3}, \mathrm{P4}\}$, and two types of consumable resources, $\{\mathrm{R1}, \mathrm{R2}\}$. There are 2 units of each type of resource.

- $\mathrm{P1}$ acquired 1 unit of R2 and requested 1 unit of R1.
- $\mathrm{P2}$ acquired 1 unit of R1.
- $\mathrm{P3}$ acquired 1 unit of R1 and requested 1 unit of R2.
- $\mathrm{P4}$ acquired 1 unit of R2.

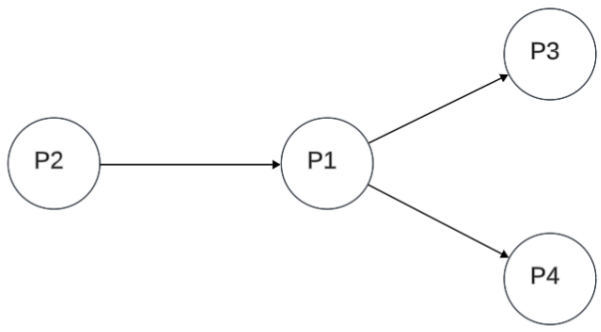
Which of the following processes are NOT in deadlock?

A. $\mathrm{P1}$
B. $\mathrm{P2}$
C. $\mathrm{P3}$
D. $\mathrm{P4}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

In a uniprocessor machine one process can run at a time, in such a machine four threads $\mathrm{P1}, \mathrm{P2}, \mathrm{P3}$ and $\mathrm{P4}$ of a process P are running concurrently, sharing a critical section. The critical section has some property that every thread must enter in their critical section according to the given topological order of their dependency.



The minimum number of binary semaphores required to maintain this dependency?

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a process Z with three threads A, B, and C. A single semaphore is to be used for synchronization among these threads. For which of the following synchronization scenarios would it be appropriate to use a single semaphore initialized to the value 2 ?

A. Ensuring that Thread A completes before Thread B.
B. Preventing more than 2 of the threads from running function $f()$ simultaneously.
C. Ensuring that Thread A runs after Thread B and Thread C have both completed.
D. Preventing Thread A or B from running function $f()$ simultaneously.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements are FALSE about threading?

A. Threads share an address space with other threads in the same process.
B. Threads share register state and stack in the same process.
C. The kernel can only create threads equal to the number of processors in the system.
D. Switching between threads of a process requires flushing of TLB.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following problems are associated with spin locks?

A. Deadlock
B. Starvation due to lack of fairness
C. Priority inversion cannot occur
D. Context-switch overhead is eliminated

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Measurements of a certain system have shown that the average process runs for a time T before blocking on $\mathrm{I/O}$. A context switch(switching to a ready process after the termination of another process is also considered a context switch) requires a time S , which is effectively wasted. For round-robin scheduling with quantum Q, if the efficiency of cpu is 50% then which of the following is true?

A. $\mathrm{Q} > \mathrm{T}$
B. $\mathrm{S} < \mathrm{Q} < \mathrm{T}$
C. $Q = S$
D. Q nearly 0

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Suppose that the following processes arrive for execution at the times indicated. Each process will run the listed amount of time.

Processes	Arrival Time (sec)	Running Time (sec)
$P1$	0.0	16
$P2$	0.3	5
$P3$	1.0	3

What will be the average turnaround time if the shortest remaining first scheduling is used(roundoff to 2 decimal places)?

Your Answer: | Correct Answer: 11.67 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a system with 16-bit virtual address space and a single-level page table. Page size is 16 Byte and each page-table entry (PTE) occupies 1 byte. The snapshot of the main memory for the process is given below:

Main Memory:	Page 16	.	.	Page 512	.	.	Page 255	.	.	Page 64
	Frame 0			Frame 67			Frame 1000			Frame 4095

i. virtual address: 8192
ii. virtual address: 4095

What are the physical addresses corresponding to the given virtual addresses?

A. (i) 256, (ii) 16015
B. (i) 1072, (ii) 65535
C. (i) 256, (ii) 65535
D. (ii) 1072, (ii) 16015

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A computer system provides each process a maximum of 256 TeraBytes(1 TeraByte $= 2^{40}$ bytes) of address space and a physical memory of 4GB. The system uses paging with a page size of 8 KB. Assuming a single-level linear page table is implemented to map virtual pages to physical frames, and b-bit are required for the indexing in the page table. Each entry of the page table uses 8 bytes. All the pages in the system have the same size. The value of b is _____ . (Answer in integer)

Your Answer: | Correct Answer: 35 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A computer system has N bytes of physical memory and M bytes of virtual address space. The page table(single-level) of a process contains Q entries, and each page-table entry (PTE) requires exactly K bytes. Assume that the page table stores only the frame number. What is the size of a page in this system?

$$\begin{aligned} \text{Physical address space} &= \text{Number of frames} * \text{page size} \\ \text{Virtual address space} &= \text{Number of pages} * \text{page size} \end{aligned}$$

A. $\mathrm{N/Q}$
B. $\mathrm{M/Q}$
C. $N/Q * K$
D. $\mathrm{M/Q^* K}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A computer system uses a 48-bit virtual address space and a 32-bit physical address space. The page table is currently organized as a three-level page table, with 4 KB page size. The size of page table entry (PTE) at all levels is 1 byte, and all pages are of equal size.
The system designers intend to reduce the page table structure from 3 levels to 2levels to reduce the number of memory accesses in address translation. What should be the maximum page size (in KB) for such page table organisation?

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

1. Trap to the operating system.
2. Check that the page reference was legal, and determine the location of the page in secondary storage and issue a read request.
3. Save the registers and process state.
4. allocate the CPU core to some other process(while blocked).

5. Receive an interrupt from the storage $\mathrm{I/O}$ subsystem (I/O completed).
6. Correct the page table and other tables to show that the desired page is now in memory.
7. Save the registers and process state for the other process.
8. Wait for the CPU core to be allocated to this process again.
9. Restore the registers, process state, and new page table, and then resume the interrupted instruction.

Which of the following is the correct order of execution for the fault service?

A. $1, 2, 3, 4, 5, 6, 7, 8, 9$
B. $1, 3, 2, 4, 5, 7, 6, 8, 9$
C. $1, 2, 3, 5, 4, 7, 6, 8, 9$
D. $1, 3, 2, 4, 5, 6, 7, 8, 9$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A unix file system has 2 KB blocks and 4 byte disk addresses. Each i-node contains 10 direct entries, one singly-indirect entry and one doubly-indirect entry. The space occupied by the i-node in this file system if file descriptors and other administrative information take 1 KB of space is _____ ?

A. 2048 Bytes
B. 1048 Bytes
C. 2088 Bytes
D. 1072 Bytes

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a memory system with a TLB access time of 10 ns and a memory access time of 110 ns - assume the memory access time includes the time to check the TLB. If the effective access time is 10% greater than the TLB access time, what is the hit ratio H?

Your Answer: | Correct Answer: 0.99 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a swapping system in which memory consists of the following hole sizes in order:

$$10\text{MB}, 4\text{MB}, 20\text{MB}, 18\text{MB}, 7\text{MB}, 9\text{MB}, 12\text{MB}, 15\text{MB}$$

Successive segment requests of sizes $12\text{MB}, 10\text{MB}, 9\text{MB}$ arrive for allocation. If the Next-Fit allocation strategy is used, the memory holes selected for these requests are:

A. $12\text{MB}, 15\text{MB}, 10\text{MB}$
B. $20\text{MB}, 12\text{MB}, 15\text{MB}$
C. $12\text{MB}, 10\text{MB}, 9\text{MB}$
D. $20\text{MB}, 18\text{MB}, 9\text{MB}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a virtual memory system with TLB that can hold 1024 entries and can be accessed in 1 ns, page table entries can be found in 100 ns in main-memory and the average page replacement time is 6 ms. If page references are handled by the TLB 99% of the time, and only 0.01% lead to a page fault, what is the effective address-translation time(in nanoseconds, round of 2 decimal places)?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #19 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Consider in a system the following page replacement policies are used:

- FIFO (First-In First-Out): Evicts the page that entered memory earliest.
- LRU (Least Recently Used): Evicts the page that was least recently referenced.
- OPT (Optimal): Evicts the page that will not be used for the longest time in the future.

Which of the following statements is/are correct?

A. LRU can never perform worse than FIFO for the same reference string and number of page frames.
B. LRU and FIFO perform the same when all referenced pages are distinct.
C. FIFO suffers from Belady's anomaly, but LRU and OPT do not.
D. Increasing the number of frames in FIFO incurs the same page misses as of LRU.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are True about the Bankers algorithm?

A. Banker's algorithm implements the deadlock avoidance strategy.
B. If a resource request is not approved and the resource request is still processed, then the system may enter into deadlock.
C. A system always causes deadlock if no safe sequence exists.
D. A process having request >= available resources is always granted by Banker's algorithm.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a system with 32-bit virtual address space with 64-Byte page frames mapping to a physical memory having 1 M words. The word length is 32 bits. A 4-way set associative TLB of 256 words is used to optimize the page access time. The number of virtual address space that can be accessed without a page table walk, if the entire TLB is fully utilized?

A. 2^6
B. 2^8
C. 2^9
D. 2^{16}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A unix file system has 2 KB blocks and 4 byte disk addresses. Each i-node contains 10 direct entries, one singly-indirect entry and one doubly-indirect entry. Suppose half of the files are exactly 1.5 KB and the other half of all files are exactly $2\mathrm{KB}$, what percentage fraction of disk space would be wasted?

Your Answer: | Correct Answer: 12.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a system that uses 2-level page tables for the processes. Page tables for both levels are in the main-memory. The system has a TLB with 90% hit ratio. The Access time of TLB is 2 ns and main-memory access time is 50 ns . For a certain process the page table has 10% dirty pages that require page transfer from secondary storage to main-memory. The page service time is 1000 ns. The average time(in ns) taken to translate virtual address to physical address ?

Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following C program:

```
int x[N];
int step = M;

while(1){
    for( int i = 0; i < N; i += step){
        x[i]  = x[i] + 1;
    }
}
```

Assume

- `int` occupies 4 bytes,
- page size $= 1$ KB (1024 bytes),
- TLB has 16 entries, and the TLB uses LRU replacement policy.

If M and N (positive integers) are chosen such that the program suffers a TLB miss on every execution of the inner loop (i.e., every reference $x[i]$ in one full for loop iteration sequence causes a TLB miss on every run of the for loop), then the minimum value of $(\mathrm{M}+\mathrm{N})$ is .

Your Answer: | Correct Answer: 4353 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a system with K resources of the same type, and each processes is requesting for $\mathrm{m}, (\mathrm{K} > \mathrm{m})$ number of resources. What could be the maximum number of processes such that the system is in safe state?

A. $\leq \left\lfloor \frac{K-1}{m-1} \right\rfloor$
B. $\leq K-1$
C. $\leq \left\lfloor \frac{K-1}{m} \right\rfloor$
D. $\leq K-m+1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: S1 and S2 both initialised to 1. The threads also share a global variable x (initialized to 0). The threads execute the code shown below.

Thread T1:	Thread T2:
$\text{wait}(S1);$	$\text{wait}(S2);$
$x = x + 1;$	$\text{wait}(S1);$
$\text{print}(x);$	$x = x + 2;$
$\text{signal}(S1);$	$\text{print}(x);$
	$\text{signal}(S2);$

Which of the following outcomes is/are possible when threads execute concurrently?

A. T1 runs first and prints 1, T2 runs next and prints 3.
B. T2 runs first and prints 2, T1 runs next and prints 3.
C. T1 runs first and prints 1, T2 does not print anything (deadlock)
D. T2 runs first and prints 2, T1 does not print anything (deadlock)

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following two threads, to be run concurrently in a shared memory (all variables are shared between the two threads):

Thread A	Thread B

`for (i=0; i<5; i++) {`
`  x = x + 1;`
`}`
	`for (j=0; j<5; j++) {`
	`  x = x + 2;`
	`}`

Assume a single-processor system, that load and store are atomic, x is initialized to 0 before either thread starts, and that x must be loaded into a register before being incremented (and stored back to memory afterwards). Which of the following are the possible values after the completion of both the threads?

A. 11
B. 12
C. 14
D. 15

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following two threads, to be run concurrently in a shared memory (all variables are shared between the two threads):

Thread A	Thread B

`for (i=0; i<5; i++) {`
`  x = x + 1;`
`}`
	`for (j=0; j<5; j++) {`
	`  x = x + 2;`
	`}`

Assume a single-processor system, that load and store are atomic, x is initialized to 0 before either thread starts, and that x must be loaded into a register before being incremented (and stored back to memory afterwards). Suppose we replace ' $\mathrm{x} = \mathrm{x} + 2$ ' in Thread B with an atomic double increment operation atomicIncr2(x) then Which of the following are the possible values after the completion of both the threads?

A. 11
B. 12
C. 14
D. 15

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A computer has 4 GB of RAM of which the operating system occupies 512 MB. The processes are all 256 MB and are independent of each other. If the CPU utilization is 99%, then what is the maximum I/O probability that can be tolerated?

A. 1%
B. 50%
C. 72%
D. 85%

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Assume a system with following specification:
1. A two set associative TLB
2. A TLB with 8 total entries
3. 2^8 byte page size
4. 2^{16} bytes of virtual memory

TLB			
Index	Tag	Frame Number	Valid
0	0x13	0x30	1
	0x34	0x58	0
1	0x1F	0x80	0
	0x2A	0x72	1
2	0x1F	0x95	1
	0x20	0xAA	0
3	0x3F	0x20	1
	0x3E	0xFF	0

Virtual Address	Physical Address
0x7E85	(i)
0xD040	(ii)
(iii)	0x3020

Page Table

Which of the flowing addresses will take place in missing entries of the above page table (Invalid: invalid entry in TLB)?

A. (i) invalid, (ii) Invalid, (iii) $0x1320$
B. (i) invalid, (ii) $0\text{x}3440$, (iii) $0\text{x}4\text{C}20$
C. (i) $0\text{x}9585$, (ii) Invalid, (iii) $0\text{x}1320$
D. (i) $0\text{x}9585$, (ii) Invalid, (iii) $0\text{x}4C20$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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